Test and Truck Data Signal Processing

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1 Recursive Least Squares with Change Detection

The goal of the present work is to remove the high-frequency measurement noise from the test and truck measurement signals of concentrations, temperature, flow-rate and urea injection rate. This signals while slowly varying, undergo abrupt changes due to the nonlinearity of the system dynamics. Thus, the problem becomes rejecting the high-frequency noise while retaining the signal jumps. This is achieved by using a recursive least-squares filter with forgetting factor which is set to zero for one sample based on the detection of the jump in magnitude. The Cumulative Sum (CUMSUM) algorithm, a special case of Sequential Probability Ratio Test (SPRT) is used for detecting the signal jumps. The algorithm and the tuning method is described in the rest of this section. Let θ be the signal to be estimated, y be the noisy measurement signal, ε be the noise. We have the signal model:

$$
\begin{aligned}
y_t &= \theta_t + \varepsilon_t \\
\hat{\theta}_t &= \lambda \hat{\theta}_{t-1} + (1-\lambda) y_t \\
\epsilon_t &= y_t - \hat{\theta}_{t-1} \\
s_t^{(1)} &= \epsilon_t \\
s_2^{(2)} &= -\epsilon_t \\
g_t^{(1)} &= \max\left(g_{t-1}^{(1)} + s_t^{(1)} - \nu, 0\right)
\end{aligned}
$$

$$
\begin{aligned}
\text{Alarm if } g_t^{(1)} > h \text{ or } g_t^{(2)} > h \\
\text{After alarm, } g_t^{(1)} = 0,\ g_t^{(2)} = 0 \text{ and } \hat{\theta}_t = y_t
\end{aligned}
$$

$$
\begin{aligned}
\textbf{\textit{Design Parameters}} &: \lambda, \nu, h \\
\textbf{\textit{Output}} &: \hat{\theta}_t
\end{aligned}
$$

Algorithm 1: Recursive Least Squares with Change Detection

1.1 Tuning of CUMSUM-RLS filtering algorithm

Start with a very large threshold h. Choose ν to be one half of the expected change, or adjust ν such that $g_t = 0$ more than 50% of the time. Then set the threshold so the required number of false alarms (this can be done automatically) or delay for detection is obtained.

- If faster detection is sought, try to decrease ν.
- If fewer false alarms are wanted, try to increase ν.
- If there is a subset of the change times that does not make sense, try to increase ν.

1.2 Equivalent time constant approximation for forgetting factor

The forgetting factor in RLS makes it behave like a low-pass filter. The equivalent time constant for the forgetting factor is given by:

$$
\begin{aligned}
& e^{-h/T_f} = \lambda \\
\implies & T_f = \frac{-h}{\ln(\lambda)} \approx \frac{h}{(1-\lambda)} = \begin{cases} 100h & \text{for } \lambda = 0.99 \\ 10h & \text{for } \lambda = 0.9 \end{cases}
\end{aligned}
$$

where, h is the sampling period and T_f is the equivalent time constant for the forgetting factor. It is thus clear that with a forgetting factor of $\lambda = 0.99$ approximately 100 to 400 immediately previous samples are used by the current parameter estimation. If $\lambda = 0.9$ that number goes down to 10 to 40.

Part I

Test Data Signal Processing

2 Frequency Response of Individual Signals

3 Results

Part II

Truck Data Signal Processing

4 Frequency Response of Individual Signals

5 Results